

- 10 Jade is a parkour enthusiast who enjoys jumping between built structures or obstacles. One day, Jade decides to jump from one 2-metre-high structure to another of the same height. The path of Jade's jump can be modeled by the quadratic equation:

$$h = -\frac{5}{2}t^2 + 5t + 1 \text{ where } h \text{ is the height in metres above the ground after } t \text{ seconds.}$$

- (a) Rewrite h in the form $b - k(t - a)^2$. [3]

$$\begin{aligned} h &= -\frac{5}{2}t^2 + 5t + 1 \\ &= -\frac{5}{2}[t^2 - 2t] + 1 \quad (M1) \\ &= -\frac{5}{2}[(t-1)^2 - (1)^2] + 1 \quad (M1) \\ &= -\frac{5}{2}(t-1)^2 + \frac{5}{2} + 1 \\ &= -\frac{5}{2}(t-1)^2 + 3\frac{1}{2} \\ &= 3\frac{1}{2} - \frac{5}{2}(t-1)^2 \quad (A1) \end{aligned}$$

- (b) Hence determine

- (i) the maximum height that Jade reaches during the jump, [1]

$$= 3\frac{1}{2} \text{ m} \quad (B1)$$

- (ii) the time at which this maximum height occurs. [1]

$$= 1 \text{ s} \quad (B1)$$

- (c) Find the total time Jade spends in the air before landing on the other building. [3]

$$\text{When } h = 2 \text{ m}$$

$$-\frac{5}{2}(t-1)^2 + 3\frac{1}{2} = 2$$

$$-\frac{5}{2}(t-1)^2 = -\frac{3}{2}$$

$$5(t-1)^2 = 3 \quad (M1)$$

$$(t-1)^2 = \frac{3}{5}$$

$$t-1 = \sqrt{\frac{3}{5}} \quad \text{or} \quad t-1 = -\sqrt{\frac{3}{5}} \quad (M1)$$

$$t = 1.774 \dots \quad t = \cancel{1.5485} \dots 0.2254 \dots$$

$$\begin{aligned} \therefore \text{total distance} &= 1.774 - \cancel{1.5485} 0.2254 \\ &= 1.5485 \dots \\ &= 1.55 \text{ s (3 sf)} \quad (A1) \end{aligned}$$